MIAE 215 Quiz #2

Please write the following programs using your compiler or a text editor. Then submit the cpp file for your program on Moodle. Don't submit any other files. You should also check your cpp file before submitting it using a text editor such as Notepad++.

If you can't find your cpp file then you should look for it in all folders of your project. You could also save the file from your compiler / IDE to the desktop and look for it there. Worst case scenario you can cut and paste your program into another text file.

You have from 5:45 pm to 6:45 pm (60 minutes) to complete the quiz and another 5 minutes to submit.

All submissions are due at 6:50 pm (except ACSD students). Late quizzes will not be accepted.

If you are not sure about a question, please state your assumptions / interpretation and proceed to solve the problem based on your assumptions.

Write a program that:

a) Inputs an integer n from the keyboard where n<=100. If n is out of range then print out an error message and ask for another input. This process repeats until a valid value for n is obtained.

b) Inputs two 1D arrays of doubles A and B (of size n) from the keyboard.

c) Inputs an integer k (from 1 to 3) from the keyboard.

d) If k = 1 then it calculates the average value of A[i] and prints it to the screen.

e) If k = 2 then it calculates the minimum value of B[i] and prints it to the screen.

f) If k = 3 then it calculates the dot product of A and B and prints the result to the screen.

g) If k is out of range then the program prints out an error message and terminates the program.

h) Parts c) to g) repeat until every valid value of k (ie menu item) has been selected at least once (or part g terminates the program). If you don't know how to do this part you can just assume parts c) to g) repeat indefinitely.